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Student Number



2018

HIGHER
SCHOOL
CERTIFICATE

Mathematics Extension 1 Task 3

General

Instructions

- Reading time – 5 minutes.
- Working time – 60 minutes.
- Write using blue or black pen
- NESA approved calculators may be used
- A formulae and data sheet is provided
- For questions 5 – 7, Show relevant mathematical reasoning and/or calculation

Section I – 4 marks (pages 2–3) • Attempt Questions 1 – 4 • Allow about 5 minutes for this section	Multiple Choice Questions	/4
	Question 5	/11
	Question 6	/13
	Question 7	/12
Section II – 37 marks (pages 4 – 7) • Attempt Questions 5 – 7 • Allow about 55 minutes for this section	Total	/40

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 20% of the Higher School Certificate Course Assessment

Section I

4 marks

Attempt Questions 1 – 4

Allow about 5 – 7 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 4 (Detach from paper)

- 1** In how many different ways can the letters of the word ‘OPTICAL’ be arranged so that the vowels are always together?
- (A) 610
(B) 720
(C) 825
(D) 920
- 2** At Hobbit school, there are six periods in each working day. In how many ways can one organise five subjects such that each subject is allowed at least one period?
- (A) 1800
(B) 3200
(C) 3600
(D) 4150
- 3** The derivative of $y = x^2 \sin^{-1}(2x)$ is:
- (A) $2x \sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-4x^2}}$
(B) $2x \sin^{-1}(2x) + \frac{x^2}{\sqrt{1-4x^2}}$
(C) $2x \sin^{-1}(2x) - \frac{4x^2}{\sqrt{1-4x^2}}$
(D) $2x \sin^{-1}(2x) + \frac{x^2}{\sqrt{1-2x^2}}$

Section 1 continues on page 3

- 4 If a, b, c are consecutive terms of an arithmetic sequence as well as a geometric sequence, then:

- (A) $a = b \neq c$
- (B) $a \neq b = c$
- (C) $a \neq b \neq c$
- (D) $a = b = c$

End of Multiple choice section

Section II

36 marks

Attempt Questions 5 – 7

Allow about 55 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 5 – 7, your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (11 marks) Use a SEPARATE writing booklet

	Marks
a) Find the derivative of $\sin^{-1}(\cos x)$ for $\pi < x < 2\pi$	2
b) Find the exact value of $\cos(2 \tan^{-1} 2)$ without the use of a calculator.	2
c) The sum of the first ten terms of the series $\log_2 \frac{1}{x} + \log_2 \frac{1}{x^2} + \log_2 \frac{1}{x^3} + \dots \dots \dots (x > 0) \text{ is } -440.$ Find the value of x .	2
d) Consider the series $\sin^2 x + \sin^4 x + \sin^6 x + \dots \dots \dots \dots \dots 0 < x < \frac{\pi}{2}.$ Show that the limiting sum exists	2
e) A committee of five is to be formed from 4 Liberal senators, 3 Labor senators and 2 Democrat senators.	
(i) How many different committees can be formed that have 3 Liberals, 1 Labor and 1 Democrat?	1
(ii) If the committee is to be chosen at random, what is the probability that there will be a Liberal majority in the committee?	2

End of Question 5

Question 6 (13 marks) Use a SEPARATE writing booklet

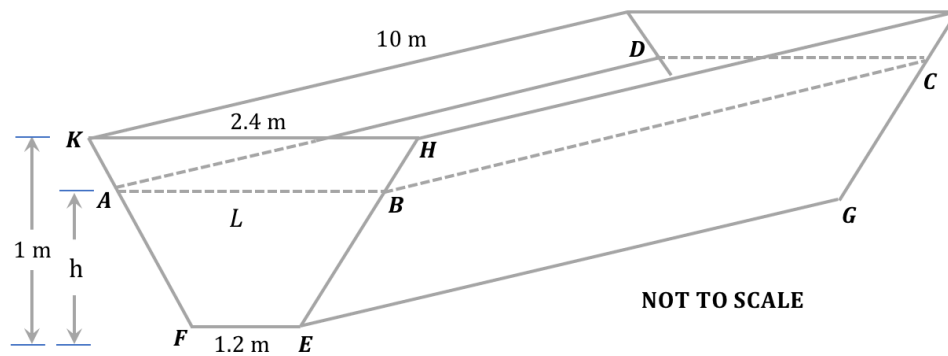
Marks

- a) Eight people attend a meeting. They are provided with two circular tables, one seating 3 people and the other seating 5 people.
- (i) How many seating arrangements are possible? **2**
- (ii) If the seating is done randomly, what is the probability that a particular couple are on different tables? **2**
- b) Andy played a game against the computer.
- Andy gained 400 points on the first day he played. On each subsequent day, he gained 90% of the points he gained on the previous day.
- Find the minimum number of days for Andy to gain at least 95% of the theoretical maximum points he can earn. **3**
- c) Betty has one black, one yellow, two red, two green and three blue beads. The nine beads are arranged randomly in a line. Find the probability that:
- (i) The two red beads are next to each other. **2**
- (ii) All red and green beads are separated. **2**
- (iii) Either the two red beads are next to each other or the two green beads are next to each other or both. **2**

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet

Marks



- a) The trough shown contains water to a depth of h metres. The volume of water inside the trough is given by $V = 6h(h + 2)$. (*Do not prove this*)

The trough is being topped up at a rate of $0.03\text{ m}^3/\text{min}$.

- (i) Find the rate at which the water level rises when the water is 0.6 m deep. Leave your answer in exact form. 2

- (ii) Find an expression for L , the length of AB . 2

Hence show that the area of the water surface $ABCD$ is given by

$$S = 12h + 12$$

- (iii) Find the rate at which the area of the water surface will increase when the water inside the trough is 0.6 metres deep. 2

Question 7 continued on Page 7

Question 7 continued

- b) Let $f(x)$ be a continuously differentiable function with the inverse function denoted by $g(x)$ and $f(g(x)) = x$. [Do not prove this].

- (i) Show that

1

$$g'(x) = \frac{1}{f'(g(x))}$$

- (ii) Given that $f(x) = \sqrt{x+2}$.

2

Without finding $f^{-1}(x)$, find the derivative of $f^{-1}(x)$ using the result in part (i).

- (iii) Find the domain of $f^{-1}(x)$.

1

- (iv) Find the points of intersection $f(x)$ and $f^{-1}(x)$

2

End of Paper

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Student Number

Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 4. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

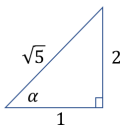
If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 ↑ *correct*

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐

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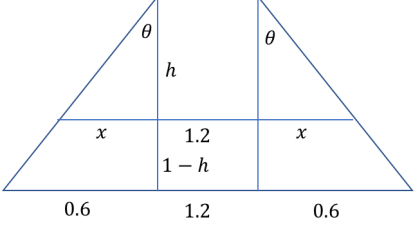
1.	3 vowels and 4 consonants No. of ways = $3! \times 5! = 720$.	B	
2	5 subjects can be organised in 6 subjects in 6P_5 Ways. Any of the 5 subjects can be organised in the remaining period in 5C_1 ways. Two subjects are alike in each of the arrangement. So, divide by $2!$. Number of arrangements = $\frac{{}^6P_5 \times {}^5P_1}{2!} = 1800$	A	
3	$y = x^2 \sin^{-1}(2x)$ $\frac{dy}{dx} = x^2 \cdot \frac{1 \times 2}{\sqrt{1-4x^2}} + 2x \sin^{-1}(2x)$ $= \frac{2x^2}{\sqrt{1-4x^2}} + 2x \sin^{-1}(2x)$	A	
4	$b = \frac{a+c}{2}$ $b^2 = ac$ $\left(\frac{a+c}{2}\right)^2 = ac$ $(a+c)^2 = 4ac$ $(a-c)^2 = 0$ $\therefore a = c. \quad \therefore b = \frac{c+c}{2} = c$ $a = b = c$	D	
Q5(a)	$\frac{d}{dx} \sin^{-1}(\cos x) = \frac{1}{\sqrt{1-\cos^2 x}} \cdot (-\sin x)$ $= \frac{1}{\sqrt{\sin^2 x}} \cdot (-\sin x). \quad \mathbf{1 \text{ mark}}$ $= \frac{\sin x}{ \sin x } = 1 \text{ for } \pi < x < 2\pi. \quad \mathbf{1 \text{ mark}}$	1 mark: correct differentiation 1 mark: correct working and reasoning	
b)	Let $\alpha = \tan^{-1} 2$ $\therefore \tan \alpha = 2$  $\sin \alpha = \frac{2}{\sqrt{5}} \quad \text{and} \quad \cos \alpha = \frac{1}{\sqrt{5}}$ $\cos(2 \tan^{-1} 2) = \cos^2 \alpha - \sin^2 \alpha$ $= \left(\frac{1}{\sqrt{5}}\right)^2 - \left(\frac{2}{\sqrt{5}}\right)^2 = -\frac{3}{5}$	1 mark: calculates correct values of $\sin \alpha$ and $\cos \alpha$ 1 mark: calculates $\cos 2\theta$ using their $\sin \alpha$ and $\cos \alpha$	

c)	$\log_2 \frac{1}{x} + \log_2 \frac{1}{x^2} + \log_2 \frac{1}{x^3} + \dots \dots (x > 0)$ $= -\log_2 x - 2\log_2 x - 3\log_2 x - \dots$ $S_{10} = -440$ $= -\log_2 x (1 + 2 + 3 + \dots + 10) = -440. \quad \mathbf{1 \text{ mark}}$ $= -\log_2 x \left(\frac{10}{2} (1 + 10) \right) = -440$ $= -\log_2 x \times 55 = -440$ $= \log_2 x = 8$ $x = 2^8 = 256. \quad \mathbf{1 \text{ mark}}$	1 mark: produces the series 1 mark: calculates the sum of the series and hence, the value of x.	
d)	For limiting sum to exist, it needs to be proved that $r < 1$ $\sin^2 x < 1$ to be proved. For $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ Hence, $0 < \sin^2 x < 1$ ie. $r < 1$ Limiting sum exists. (It is a convergent series)	1 mark: explains that for $0 < x < \frac{\pi}{2}$, $0 < \sin x < 1$ 1 mark: proves $r < 1$	
(e)(i)	No. of committees = $4_{C_3} \times 3_{C_1} \times 2_{C_1} = 24$	1 mark: correct answer with working	
(ii)	For Liberal majority, 4 Lib, 1 Lab = $4_{C_4} \times 3_{C_1} = 3$ 4 Lib, 1 Dem = $4_{C_4} \times 2_{C_1} = 2$ 3 Lib, 1 Lab, 1 Dem. = 24 from (i) 3 Lib, 2 Lab = $4_{C_3} \times 3_{C_2} = 12$ 3 Lib, 2 Dem = $4_{C_3} \times 2_{C_2} = 4$ Total no. of committees = $3 + 2 + 24 + 12 + 4 = 45$ P(Lib majority) = $\frac{45}{9_{C_5}} = \frac{5}{14}$	2 marks: correct answer from correct working 1 mark: gives either total number of committees correctly or calculates the correct probability	

Question 6			
(a)(i)	Choose 5 people for the first table, 8_{C_5} ways; arrange them around the table $4!$ Ways; then the rest of the 3 people can be arranged in $2!$ Ways. (5) (3) Number of arrangements = $8_{C_5} \times 4! \times 2! = 2688$	2 marks: correct answer from correct working 1 mark: correct number of arrangements of 5 and 3 people around a circle	
(ii)	A (5) B (3) or B (5) A (3) Treat them as two groups: 6 people and 2 people (A, B) Choose one from (A and B) = 2 ways; Choose 4 from 6 = 6_{C_4} ways and arrange the five people around the table $4!$ Ways; the rest of the 3 can be arranged in $2!$ Ways.	2 marks: correct answer from correct working 1 mark: significant progress towards calculating the number of arrangements and calculates probability correctly	

	$\therefore$ total number of arrangements = $2 \times 6_{C_4} \times 4! \times 2!$ $= 1440$ $P(A \text{ and } B \text{ on separate tables}) = \frac{1440}{2688} = \frac{15}{28}$		
b)	The theoretical maximum = $\frac{a}{1-r} = \frac{400}{1-0.9} = 4000$ points $S_n \geq 0.95 \times S_\infty$ $\frac{400(1-0.9^n)}{1-0.9} \geq 0.95 \times \frac{400}{1-0.9} \quad 1 \text{ mark}$ $1 - 0.9^n \geq 0.95.$ $0.9^n \leq 0.05$ $n \geq \frac{\ln 0.05}{\ln 0.9} = 28.4 \dots \dots$ Minimum no. of days. = 29. 1 mark (using their S_∞)	1 mark: correctly calculates the theoretical maximum points 1 mark: writes the equation to represent the model 1 mark: solves the equation with working and gives the correct rounded answer	
c)(i)	Number of arrangements without restrictions = $\frac{9!}{2!2!3!}$ Required probability = $\frac{8!}{2!3!} \div \frac{9!}{2!2!3!} = \frac{2}{9}$ RR - - - - -	1 mark: Total no. of arrangements with no restriction 1 mark: Calculates the probability correctly (Working required)	
(ii)	No. of ways to slot in the red beads and the green beads = $6_{C_4} \times \frac{4!}{2!2!}$ $\therefore$ Required probability $= \frac{5!}{3!} \times 6_{C_4} \times \frac{4!}{2!2!} \div \frac{9!}{2!3!3!} = \frac{5}{42}$ OR - B - Y - BL - BL - BL - P(all R and G separated) = $\frac{5!}{3!} \times \frac{6_{P_4}}{2!2!} \div \frac{9!}{2!3!3!} = \frac{5}{42}$	2 marks: correct answer from correct working 1 mark: Significant progress towards calculating probability	
(iii)	P(2 red beads together and 2 green beads together) = $\frac{7!}{3!} \div \frac{9!}{2!2!3!} = \frac{1}{18}$ Required probability = $P(A) + P(B) - P(A \cap B)$ $= \frac{2}{9} + \frac{2}{9} - \frac{1}{18} = \frac{7}{18}$ OR P(2 R together, G not) = $\frac{6! \times 7_{C_2}}{3!} \div \frac{9!}{2!2!3!} = \frac{1}{6}$ P(2G together, R not) = $\frac{6! \times 7_{C_2}}{3!} \div \frac{9!}{2!2!3!} = \frac{1}{6}$ P(2 R together and 2 G beads together) = $\frac{1}{18}$ Total probability = $\frac{1}{6} + \frac{1}{6} + \frac{1}{18} = \frac{7}{18}$	1 mark: calculates the probability of arranging the red and the green beads together 1 mark: uses the answers in (i) for probabilities of 2 red beads together (same for 2 green beads together) and calculates the probability	

Question 7

a)(i)	$V = 6h(h + 2)$ $\frac{dV}{dt} = 0.03 \text{ m}^3/\text{min}$ $V = 0.03t + C$ $V = 6h(h + 2)$ $V = 6h^2 + 12h$ $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ $\frac{dV}{dh} = 12h + 12.$ $h = 0.6, \quad \frac{dV}{dh} = 12 \times 0.6 + 12 = 19.2$ $\frac{dh}{dt} = \frac{0.03}{19.2} = 0.00156 \dots \dots$ $= \frac{1}{640} \text{ cm/min}$	1 mark: evaluates $\frac{dV}{dh}$ when $h = 0.6$ 1 mark: calculates $\frac{dh}{dt}$	
(ii)	$h = 0, L = 1.2$ when $h = 1, L = 2.4$ $L = mh + b$ $L = mh + 1.2$ $2.4 = m + 1.2 \quad \therefore m = 1.2$ $L = 1.2h + 1.2. \quad 1 \text{ mark}$ $S = 10L = 10(1.2h + 1.2)$ $S = 12h + 12. \quad 1 \text{ mark}$ OR  $\tan \theta = \frac{x}{h} = \frac{0.6}{1}$ $\therefore x = 0.6h$ $L = 1.2 + 2x$ $L = 1.2 + 2 \times 0.6$ $L = 1.2 + 1.2h$	1 mark: develops the expression for L 1 mark: expression for surface area OR 1 mark: correctly expresses x in terms of h 1 mark: Gives the correct expression for L	
(iii)	$S = 12h + 12$ $\frac{dS}{dt} = \frac{dS}{dh} \times \frac{dh}{dt}$ $= 12 \times \frac{1}{640}$ $= \frac{3}{160} \text{ m}^2/\text{min}$	1 mark: Calculates $\frac{dS}{dh}$ correctly 1 mark: differentiates and calculates $\frac{dS}{dt}$	
Q7(b) (i)	Given that $f(g(x)) = x$ $\therefore \frac{d}{dx} f(g(x)) = \frac{d}{dx} (x)$ $f'(g(x)) \cdot \frac{d}{dx} g(x) = 1$ $f'(g(x)) \cdot g'(x) = 1$ $\therefore g'(x) = \frac{1}{f'(g(x))}$	1 mark: correct proof (all lines of working must be shown)	
(ii)	$f(x) = \sqrt{x+2}, \quad g(x) = f^{-1}(x)$ $f(g(x)) = x$ $f(f^{-1}(x)) = \sqrt{f^{-1}(x)+2} = x. \quad 1 \text{ mark}$ $f'(x) = \frac{1}{2\sqrt{x+2}}$ From (i),	1 mark: identifies $\sqrt{f^{-1}(x)+2} = x$	

	$\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$ $= \frac{1}{\frac{1}{2\sqrt{f^{-1}(x)+2}}}$ $= 2\sqrt{f^{-1}(x)+2}$ $= 2x. \quad 1 \text{ mark}$	1 mark: proves $\frac{d}{dx} f^{-1}(x) = 2x$	
(iii)	Domain of $f^{-1}(x)$ = Range of $f(x)$ = All real x, $x \geq 0$	1 mark: correct answer	
(iv)	The point of intersection of $y = f(x)$ and $y = f^{-1}(x)$ $y = x$ $x = \sqrt{x+2} \quad 1 \text{ mark}$ $x^2 = x+2$ $x^2 - x - 2 = 0$ $(x+1)(x-2) = 0$ $x = -1 \text{ or } x = 2$ Since, domain of $f^{-1}(x)$ is $x \geq 0$ $\therefore x = -1$ is rejected. $\therefore$ Point of intersection is $(2, 2)$. 1 mark	1 mark: demonstrates the understanding that $f(x)$ and $f^{-1}(x)$ meet at $y = x$ 1 mark: solves for x and using the domain rejects $x = -1$, and gives the point of intersection	